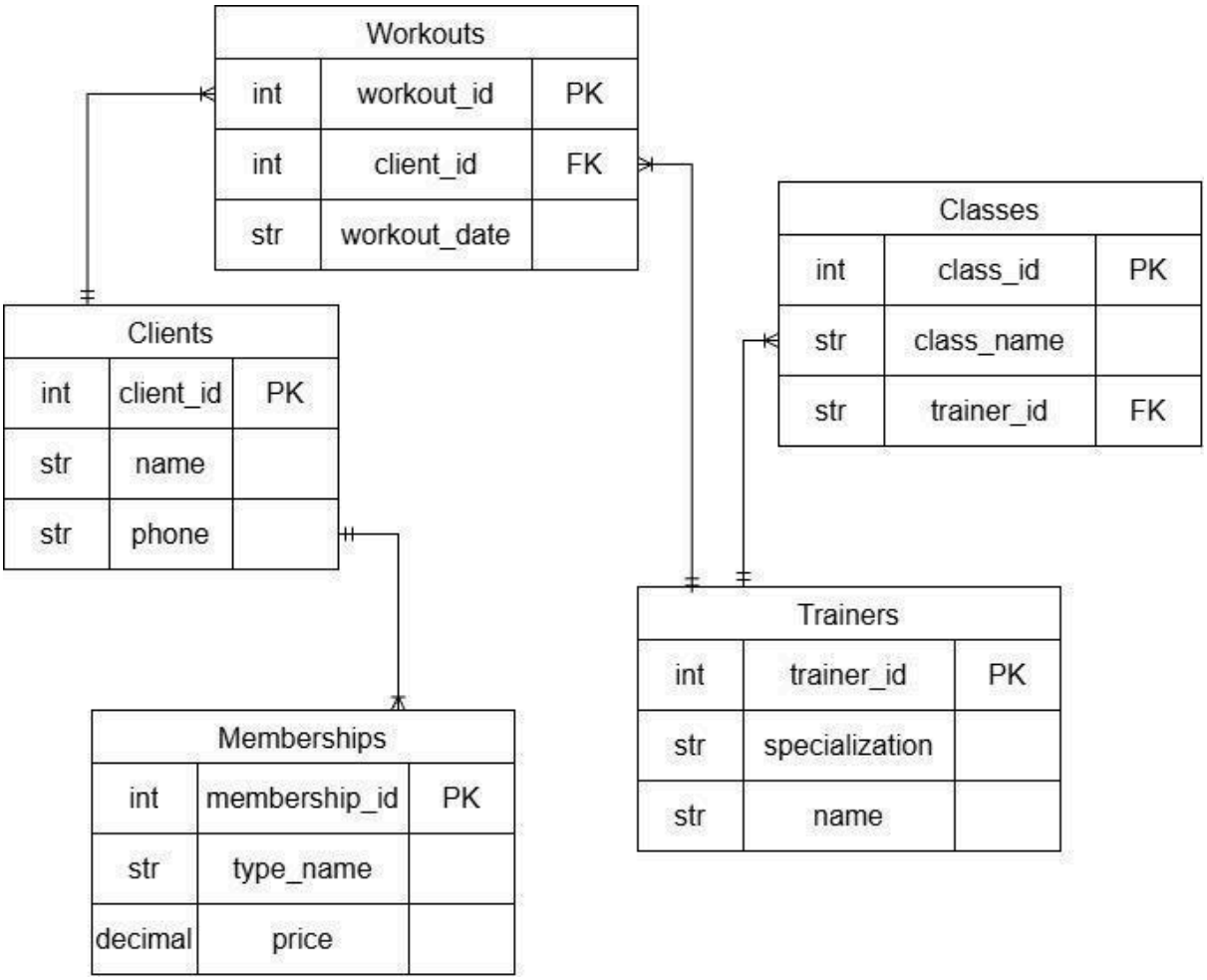
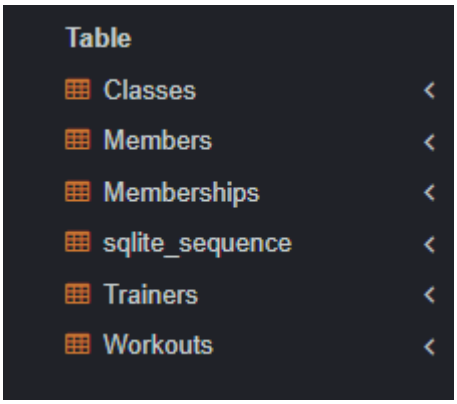
Отчёт по общему УК Шипикин Виктор Алексеевич

1. Выделил сущности, построил ER диаграмму и довёл её до 3НФ:



2. Создал БД в SQLite:



3. Написал 13 SQL запросов:

```
SELECT * FROM Trainers WHERE specialization = 'Кроссфит';

SELECT first_name, last_name, join_date FROM Members WHERE join_date > '2026-03-01';

SELECT class_name, capacity FROM Classes WHERE capacity > 14;

SELECT SUM(price) AS total_revenue FROM Memberships;

SELECT type_name, COUNT(*) AS total_sold FROM Memberships GROUP BY type_name;

SELECT member_id, AVG(duration_minutes) AS avg_duration
FROM Workouts
GROUP BY member_id
HAVING avg_duration > 50;

SELECT specialization, MAX(salary) AS max_salary FROM Trainers GROUP BY specialization;

SELECT m.first_name, m.last_name, ms.type_name
FROM Members m
JOIN Memberships ms ON m.member_id = ms.member_id;

SELECT c.class_name, t.last_name AS trainer_lastname
FROM Classes c
INNER JOIN Trainers t ON c.trainer_id = t.trainer_id;

SELECT w.workout_date, m.last_name AS client_name, t.last_name AS trainer_name
FROM Workouts w
JOIN Members m ON w.member_id = m.member_id
LEFT JOIN Trainers t ON w.trainer_id = t.trainer_id;
```

```
SELECT member_id, type_name, price
FROM Memberships
WHERE price > (SELECT AVG(price) FROM Memberships);

SELECT first_name, last_name, salary
FROM Trainers
WHERE salary = (SELECT MAX(salary) FROM Trainers);

SELECT first_name, last_name
FROM Members
WHERE member_id IN (SELECT DISTINCT member_id FROM Workouts
WHERE trainer_id IS NOT NULL);
```

4. Оформил репозиторий GitHub:

